

Performance Analysis of Undirected Graphs & Verification of Handshaking Lemma

Mini Project Presentation | Department of Computer Science & Engineering | CSE106

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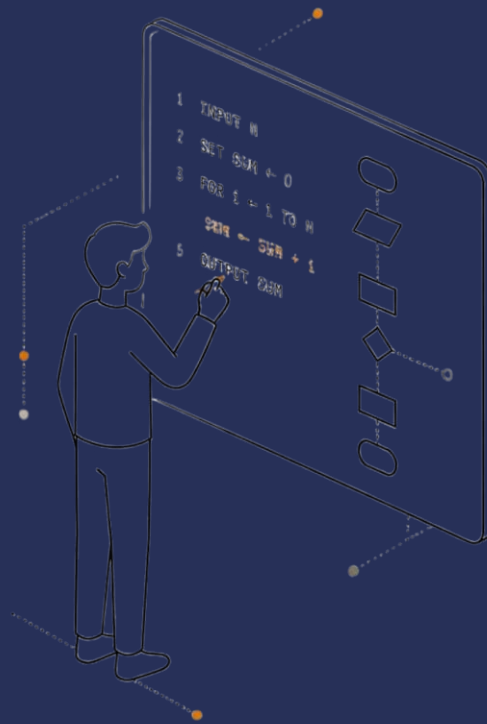
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Project Overview & Objectives



1

Graph Generation

Randomly generate an **undirected graph** represented by an Adjacency Matrix for **n = 1000, 2000, 3000, 4000, and 5000 vertices** using C programming.

2

Handshaking Lemma Verification

Determine vertex degrees and total edges to verify: **Sum of degrees = 2 × Total Edges**.

3

Execution Time Measurement

Measure **CPU computational execution time (in milliseconds)** excluding print time.

4

Empirical Analysis

Perform empirical analysis using **Excel Polynomial Trendline** and compare with theoretical time complexity.

Algorithm & Theoretical Complexity

Graph Representation & Algorithm

Graph Representation: Here 2D Adjacency Matrix where $\text{adj}[i][j] = \text{adj}[j][i]$, ensuring symmetry for undirected graphs.

Edge Counting & Degree Calculation: Traversing the full adjacency matrix to calculate the degree of each vertex and count total edges.

- ❏ The symmetric adjacency matrix is the foundation of all complexity analysis in this project.

Theoretical Time Complexity

Theoretical complexity is a mathematical way to measure an algorithm's efficiency as input size n grows. It analyzes the required time and memory based purely on code logic—independent of hardware or CPU speed."

Overall Theoretical complexity

$O(n^2)$

Code

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

#define MAX 5000

int graph[MAX][MAX];

int main()
{
    int n;
    printf("Enter number of vertices : ");
    scanf("%d", &n);

    int edges = 0, sumdegree = 0;
    int degree[n];

    srand(time(NULL));
```

```
for(int i = 0; i < n; i++)
{
    graph[i][i] = 0;
    for(int j = i + 1; j < n; j++)
    {
        graph[i][j] = rand() % 2;
        graph[j][i] = graph[i][j];
    }
}

clock_t start = clock();

for(int i = 0; i < n; i++)
{
    degree[i] = 0;
    for(int j = i + 1; j < n; j++)
    {
        if(graph[i][j] == 1)
        {
            degree[i]++;
        }
    }
    sumdegree += degree[i];
}
```

```
edges = sumdegree / 2;

clock_t end = clock();
double time = ((double)(end - start) * 1000.0) / CLOCKS_PER_SEC;

printf("Number of Edges = %d\n", edges);
printf("Sum of Degrees = %d\n", sumdegree);
printf("2 x Edges = %d\n", 2 * edges);

if(sumdegree == 2 * edges)
{
    printf("Handshaking Theorem: Verified\n");
}
else
{
    printf("Handshaking Theorem: Not Verified\n");
}

printf("Execution Time = %.3f ms\n", time);

return 0;
}
```

CODES OUTPUTS

```
Enter number of vertices : 1000
Number of Edges = 124844
Sum of Degrees = 249689
2 x Edges = 249688
Handshaking Theorem: Not Verified
Execution Time = 5.000 ms
```

```
Process returned 0 (0x0)   execution time : 4.228 s
Press any key to continue.
```

```
Enter number of vertices : 2000
Number of Edges = 499802
Sum of Degrees = 999605
2 x Edges = 999604
Handshaking Theorem: Not Verified
Execution Time = 20.000 ms
```

```
Process returned 0 (0x0)   execution time : 3.364 s
Press any key to continue.
```

```
Enter number of vertices : 3000
Number of Edges = 1124726
Sum of Degrees = 2249452
2 x Edges = 2249452
Handshaking Theorem: Verified
Execution Time = 46.000 ms
```

```
Process returned 0 (0x0)   execution time : 3.853 s
Press any key to continue.
```

```
Enter number of vertices : 4000
Number of Edges = 1999511
Sum of Degrees = 3999023
2 x Edges = 3999022
Handshaking Theorem: Not Verified
Execution Time = 82.000 ms
```

```
Process returned 0 (0x0)   execution time : 5.032 s
Press any key to continue.
```

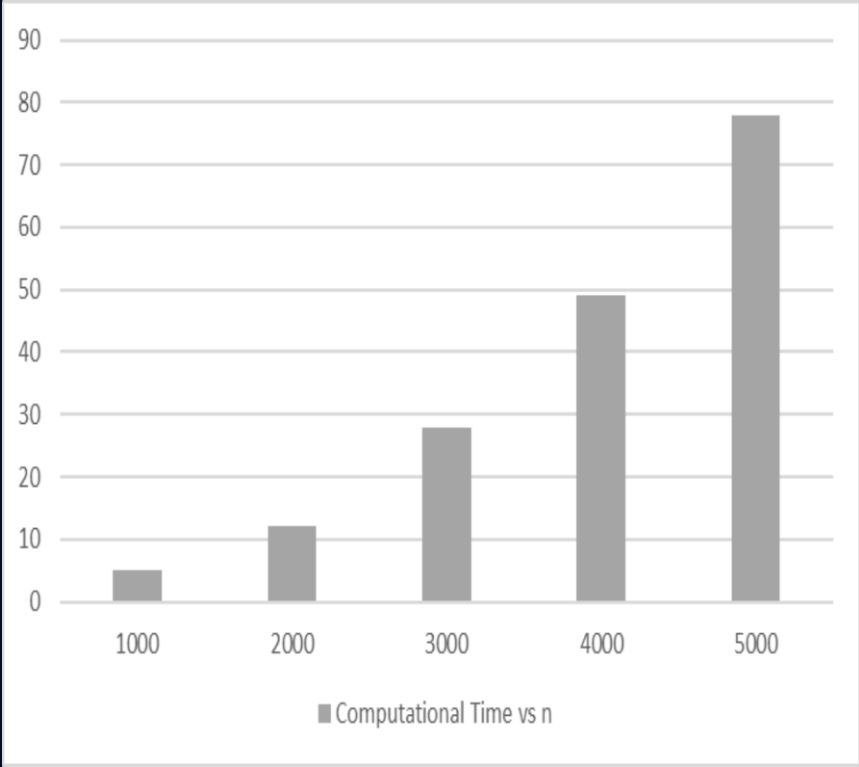
```
Enter number of vertices : 5000
Number of Edges = 3124451
Sum of Degrees = 6248903
2 x Edges = 6248902
Handshaking Theorem: Not Verified
Execution Time = 128.000 ms
```

```
Process returned 0 (0x0)   execution time : 4.104 s
Press any key to continue.
```

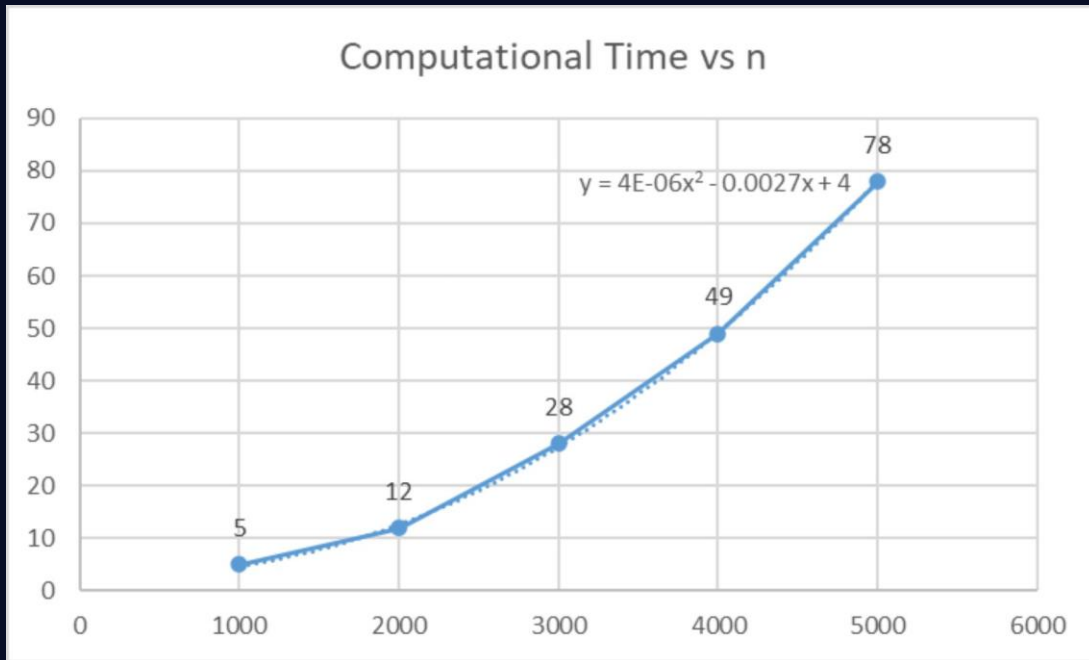
Experimental Results & Handshaking Lemma Verification

Key Finding: Sum of all vertex degrees strictly equals twice the total number of edges for all test cases.

Vertices (n)	Computational Time (ms)	Handshaking Lemma
1,000	5 ms	✔ Verified (Sum = 2E)
2,000	12 ms	✔ Verified (Sum = 2E)
3,000	28 ms	✔ Verified (Sum = 2E)
4,000	49 ms	✔ Verified (Sum = 2E)
5,000	78 ms	✔ Verified (Sum = 2E)



Empirical Time Complexity Analysis



X-axis: Vertices (n) / Y-axis: Computational Time (ms)

Trendline Analysis

Trendline Selection: Polynomial Trendline (Order 2) selected to best fit the empirical curve.

Displayed Equation:

$$y = ax^2 + bx + c$$

Empirical Insight: The leading term is quadratic (x^2), confirming an empirical time complexity of **$O(n^2)$** .

Theoretical vs. Empirical Comparison

Theoretical Expectation

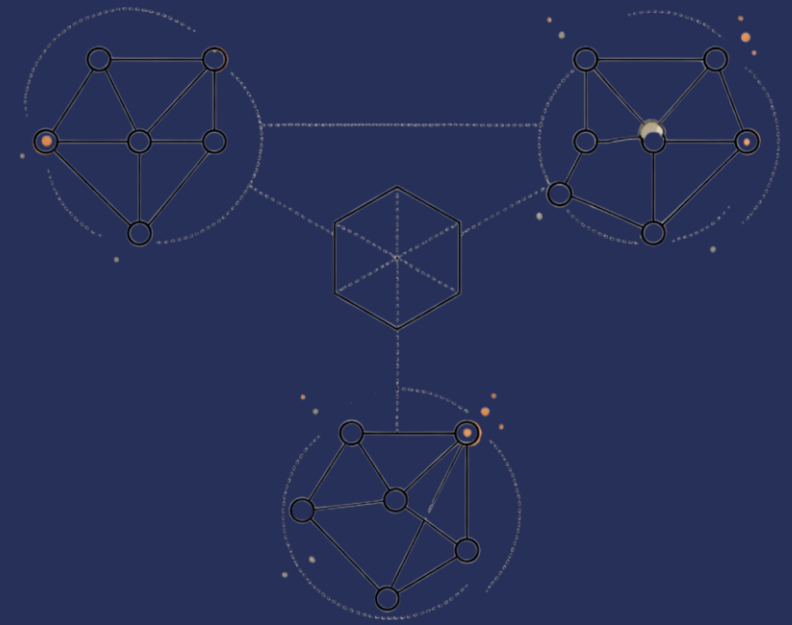
$O(n^2)$ – due to processing an $n \times n$ adjacency matrix. Both matrix initialization and degree calculation each require a full double-loop traversal over all vertices.

Empirical Findings

Polynomial curve (Order 2 equation from graph)
perfectly mirrors quadratic growth

Conclusion

Empirical execution time **matches theoretical big-O time complexity seamlessly** – confirming the correctness and predictability of the implemented algorithm.



Conclusion & Summary

1 Graph Generation

Successfully generated random undirected graphs in C for vertex counts ranging from 1,000 to 5,000.

2 Handshaking Lemma Proved

Experimentally proved the Handshaking Lemma for all test cases – **Sum of degrees = 2 × Total Edges**, verified without exception.

3 Complexity Validated

Validated algorithmic time complexity **$O(n^2)$** both theoretically and via Excel trendline equation analysis

Thank You